



CONTACT

Biel / Bienne, Switzerland
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SKILLS

Programming

Python C++ JavaScript HTML / CSS

ML & AI

PyTorch HuggingFace Contrastive Learning
Vision-Language Models Computer Vision

MLOps & Infra

W&B HPC / SLURM Docker Git Linux

Agentic & LLM

Claude API Ollama Cursor
Prompt Engineering

Web

Astro JavaScript HTML / CSS

LANGUAGES

French	Native
Tamil	Native
English	C1
German	B2
Portuguese	A2

AWARDS

BME Prize 2026

Best Master's Thesis, University of Bern

ACTIVITIES

- European finalist, Eurobot 2022 (2nd in Switzerland), competitive robotics
- Swiss jury member, Prix Goncourt de l'Université 2025
- Circus acrobatics, Théâtre de l'Heure Bleue

Nirosh Sivanesan

AI/ML Engineer

Research-to-Production

PROFILE

AI/ML engineer working at the boundary of medical imaging research and production deployment. Built physics-aware Vision-Language Models on nine international medical datasets, awarded the BME Prize 2026. Previously delivered industrial predictive maintenance systems at HEIA-FR that moved from prototype to client deployment. Most useful where ML systems need both research depth and engineering discipline.

EXPERIENCE

Research Engineer

Aug 2025 – Feb 2026

University of Bern, ARTORG Centre · Bern

- Designed a physics-aware contrastive learning framework for Vision-Language Models in medical imaging, evaluated across nine international datasets.
- Built and maintained a full HPC/SLURM training and benchmarking pipeline supporting large-scale multimodal experiments.
- Awarded the BME Prize 2026 for the originality and rigour of the research.

Research Associate

2023 – 2025

HEIA-FR, SeSi Institute · Fribourg

- Designed and deployed a digital twin pipeline automating industrial component monitoring, replacing manual inspection cycles.
- Built predictive maintenance prototypes for industrial clients; systems progressed from proof-of-concept to operational deployment.
- Engaged the full ML lifecycle: data ingestion, model training, pipeline engineering, deployment, and maintenance.

Engineering Intern

2020

Liebherr Machine SA · Bulle

- 12-month internship across production, technical office, and hydraulic and thermal R&D.

PROJECTS

- **MODIA** Hybrid physics-ML digital twins for prescriptive maintenance, validated on real industrial installations. Cuts sensor count and modelling effort while improving anomaly detection. (HEIA-FR.)
- **Physics-Aware VLM Framework** Master's thesis. Contrastive learning across nine medical imaging datasets. PyTorch, HPC/SLURM. (Private, available on request.)
- **AlphaForge** Quantitative equity pipeline: Fama-French beta decomposition, rolling volatility, FastAPI service. Dockerised, property-tested.
- **gVXR CXR Simulator** Synthetic chest X-ray augmentation pipeline using gVXR for medical imaging research. Python.

EDUCATION

MSc in Artificial Intelligence in Medicine

2023 – 2026

University of Bern

BSc in Mechanical Engineering

2019 – 2023

HEIA-FR

Bilingual FR / DE